Wearable Foot Pressure and Gait Analysis System

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***Abstract*—In this paper, the development of an insole foot pressure and gait analysis system has been presented with potential application in diabetic foot ulceration identification and adaptations for orthopedic applications. The system is a low cost, reliable method that emphasizes portability and efficiency. The system employs Force Sensitive Resistors (FSR 402, FSR 406) for pressure sensing and a six degree of freedom gyroscope, MPU 6050 for gait identification. The pressure and the gait pattern so obtained is mapped on to LabVIEW dashboard via wireless communication with ESP8266 Microcontroller.**

Keywords—Plantar Pressure, Gait, LabVIEW, Portability, in-shoe, FSR, MPU6050, ESP8266

I. Introduction

Plantar pressure measurement is one of the fastest growing topics in the field of biomedical Instrumentation. It is widely used such as biomedicine, sport biomechanics, diagnosing lower limbs, identification of diseases such as diabetic foot ulcers to being used for recovery from surgical procedures or to monitor patients that are undergoing rehabilitation[1]. Monitoring pedobarographic functions[2] enables to provide personalized healthcare to patients as well as footwear designed to each specific individual in sports[3]. Diabetes is one of the world's most widespread lifestyle diseases. It has multiple devastating effects on the body, one of which is foot ulcerations[1]. Studies have indicated diabetic patients have up to 25% more risk over a normal individual in developing foot ulcer[4]

A multitude of different pressure measuring methods can be utilized to sense the pressure that is applied by the foot, and these include capacitive sensors[5], force sensitive resistors [6], strain gauges, optical methods. The application of these individual methods can be utilized in either an in-shoe[7] or a platform pressure sensing method. The in-shoe method is gaining traction in recent times due to its lower cost, portability and overall acceptable accuracy and sensitivity.

Gait analysis is a method to identify individual movements by studying their gait cycles [8]. The application of the analysis of such movements can lay a foundation to the detection of Parkinson's disease in its early stages [9]. Lower limb gait analysis can be done using a single inertial measurement unit which provides the desired information [10].

This paper presents the design and development of a low cost four-point insole plantar pressure sensing system involving the use of force sensitive resistors and an Inertial Measurement Unit.

II. Literature Survey

The Plantar Pressure Measurement and Gait can be extracted in different methods under different conditions. Each approach has got its advantages and disadvantages. The different sensors which can be used to fetch the plantar pressure includes capacitive sensors [11], force sensitive sensors, piezoelectric sensors [12], strain gauges. For high precision applications, static optical method [13] are also seemed to be employed.

A. Pressure measurement system

The platform systems are built of stiff, flattened arrays of sensing components arranged in a matrix structure, attached to the ground, and designed to accommodate natural gait motions[14]. Ease of use is an advantage of platform systems, but individuals will have to undergo a learning curve to get used to natural gait.

In-shoe systems[7] have flexible sensors that are embedded into the insole of the shoe. Such systems are portable that provides their application in different gait studies, footwear design and terrains

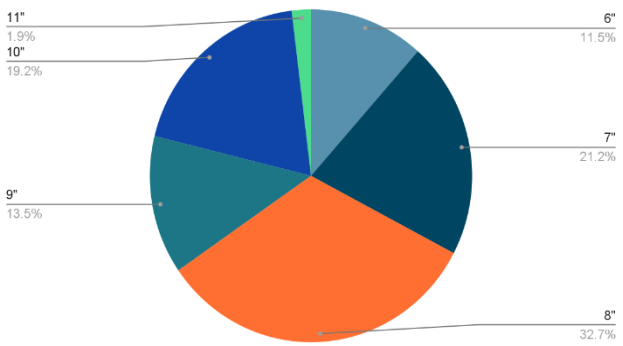

Fig 1: Survey result of the foot size

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From between the available choice of sensors, the force sensitive resistors [7] were selected due to its size, low cost, accuracy. Optical methods for pressure sensing were eliminated due to the system being an in-shoe system rather than a platform system [14]. Both piezo resistive sensors and strain gauges are sensitive to temperature variations [7] and over constant use will require calibration on a much more frequent scale. No adverse temperature effects were identified for capacitive sensors and force sensitive resistors. Load cells were also considered but were eliminated due to their thickness [5]. Due to force sensitive resistors having similar operating characteristics to capacitive sensors but being thinner, cheaper and easily available, they were selected [6].

The FSR 402 and FSR 406 [15] are part of the FSR 400 series produced by Interlink Electronics. The FSR 402 is circular sensor with 13mm*25mm while the FSR 406 is a square sensor that is 43.69mm in size [16]. With gradual increase in the force applied to the sensor surface, the force-sensitive resistors, which are polymer thick-film devices, will produce a decrease in resistance. They have applications in HMI devices such as robots, medical devices, industrial controls, and automotive electronics.

B. *Gait analysis system*

For gait analysis, the ideal sensor for identifying all the parameters of gait was chosen to be the MPU6050 [8] which is a 6 degree of freedom Inertial Measurement Unit with a built-in gyroscope, accelerometer [17][18], magneto resistive sensor[19]. The sensor has characteristics such as low power consumption, low cost and high performance. After the ideal sensor has been identified [20], the ideal position for the placement of the sensor had to be identified [21]

- Location 1: Over the central portion of *the first cuneiform*. This location is very stable during motion. This location produces the highest accuracy
- Location 2: The upper side of the foot over the *second metatarsophalangeal unit*. This part is flexible during motion. This flexibility reduces its accuracy.
- Location 3: The lateral side of the foot over the *fifth metatarsal unit.* This part is also flexible during motion and has similar accuracy to location 2.
- Location 4: The ankle joint or the region over *the talar dome.* It is a mobile part due to it being located at the ankle joint but provides sufficient accuracy.
- Location 5: The back heel of the foot or the *Achilles tendon*. This is relatively stable portion of the foot during motion and its accuracy is only second to location 1.

From these locations, location 5 was chosen as the position for the placement of the IMU sensor[6]. Even though this region is comparatively less accurate than Location 1, it provided a correct-enclosed position for the placement of the entire circuitry which includes the ESP8266, Battery, multiplexer and the IMU.

III. Survey

The production of an insole the requirement for proper sizing seriously affects the functioning of the force sensitive resistors [22]. In an event that the size of the foot is not correct for the pre-set size of the insole and the footwear then there will be a permanent compression of the FSR on the foot which will results in no variations in pressure distribution being shown in that particular sensor [15]. This problem will also affect the gait cycle as well. Therefore, it was of great importance that the ideal foot size be identified.

A survey was conducted to identify the average foot size among individuals of a particular age group. The survey was conducted among individuals of 18 to 22 years of age. This was conducted in order to develop the size of the in-sole to match the average foot size. The survey results have shown that the average foot size is 8 inches

There are multiple existing devices for the foot pressure and gait analysis in the present scenario. F-Scan, which uses insole pressure sensors is one of the leading solution for the plantar pressure measurement[33]. However, this methodology is quite expensive and it uses a proprietary software for the analysis. Paro-tech was another comparative technology which was using peixo resistive hydro cells, but it was unable to provide a real time measurement system [28]. In addition, the bulkiness, response time and slipping of the insole was found to be some of the main drawbacks from the existing solutions of insole plantar pressure measurement[32].

IV. Methodology

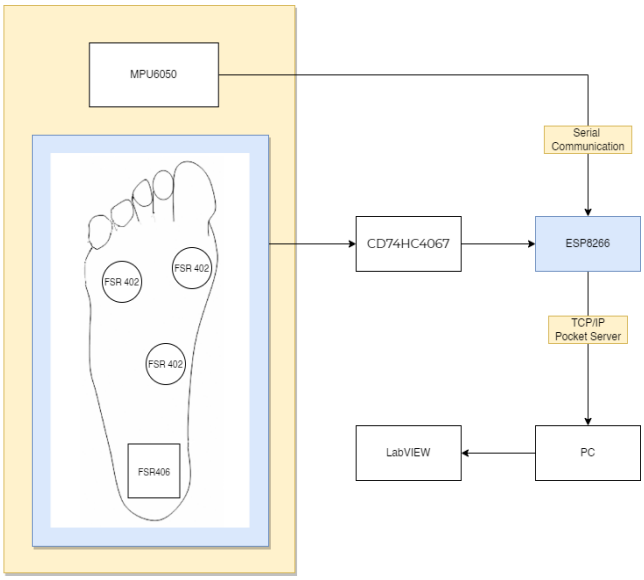

Fig 2: Block diagram representation of proposed model

Pressure is taken from the identified four sensitive points [12] on the foot using force sensitive resistors that are imprinted in a custom designed insole [23]. The sensors will gauge the conductive foam's resistance between the two electrodes when pressure is applied to the insole [6]. When

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pressure is applied, the conducting layers bend, increasing the current flowing through the resistive sensor.

Being an analog output, the four sensors need to be multiplexed with the aid of a multiplexer CD74HC4051 EC 16*1 [12]. The microcontroller unit used is the ESP8266 owing to its low cost as well as its accessibility as an open-source device [24]. It has an inbuilt TCP/IP stack including firmware that runs on the ESP8266 Wi-Fi SoC.. The multiplexer will combine 15 single analog input to the ESP8266. The 4-individual sensor outputs to the multiplexer

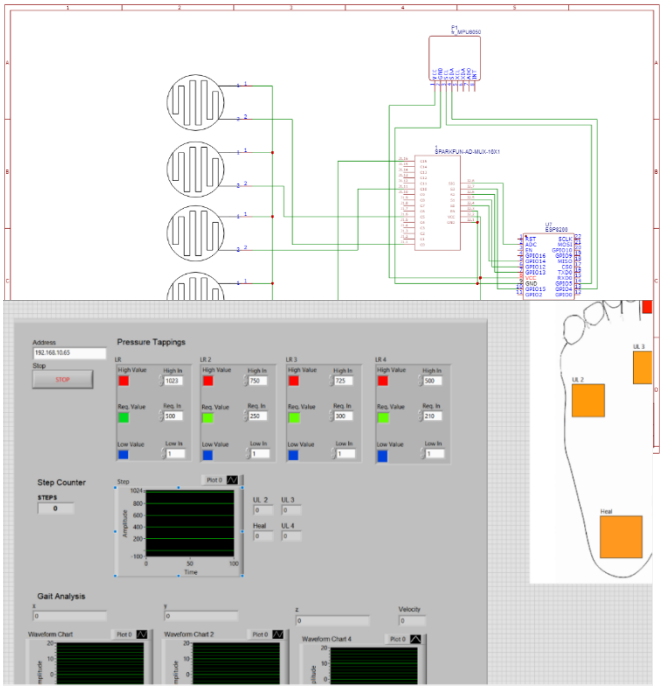

Fig 4: Front panel of the Virtual Instrument developed in LabVIEW

will be connected to 4 pins with specific addresses [25]. These individual addresses will be pointed out with an address pointer and using binary values from 0000 to 1111 to denote all possible 15 inputs and the value of the sensor will be identified. A loop will be executed that reads the values from the sensors i.e., these addresses in very short span of time. This will work with delays of milliseconds in reading and receiving the sensor value but in real time the delay will be negligible. The collected output from the sensor is fed into the ESP8266 [24]. A Virtual Instrument is developed in LabVIEW which communicates with the ESP8266 via TSP/IP communication. Under run condition, each Pedobarographic functions are gradient mapped to a custom color in the Virtual Instrument developed [26]. This is continuously analyzed to study or predict different clinical conditions including foot ulcers. When it comes to gait, a single MPU 6050, an Inertial Measurement Unit will be applied to sense the position of the leg/foot in every 2.5 µs interval.

Steps are identified as a function of the pressure exerted upon the heal. The initial position for the gait cycle was the heel down position and maximum pressure towards the end of stance phase, just before the strike of the other foot[31]. This methodology is employed for calculating the number of steps [27] taken by the individual. The pressure from this point is simultaneously plotted in the waveform chart and whenever the ESP is coded in such a way that whenever there exerts a pressure more than the cutoff, a variable is incremented by a single step. This variable is labeled as the Step Counter in the VI. The accelerations along x, y and z axis is directly obtained from the IMU Sensors which is plotted in waveform chart in the virtual instrument. The velocity was identified with the help of the following equation from the in-built accelerometer of the Inertial Measurement Unit.

$$v = u + at$$

Consider $t = 0$, $u = 0$

Let $v = at$ (for $u=0$)

We have a_x, a_y, a_z from the Gyro-accelerometer.

This gives the acceleration,

$$a = \sqrt{ax^2 + ay^2 + az^2}$$

where,

- a_x, a_y, a_z are the accelerations along x,y and z axis
- u : initial velocity of the subject
- v : final velocity of the subject

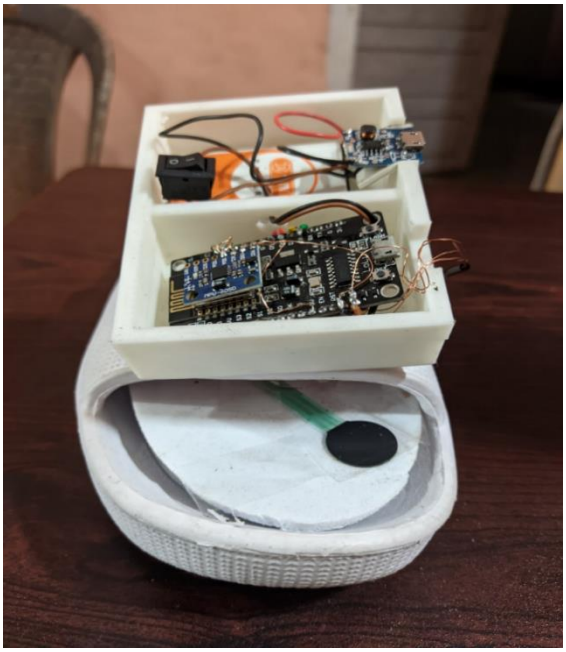

Fig 5: Hardware Prototype developed

The entire system will be powered by a lithium polymer battery with a capacity of 1020 mAh which provides roughly two hours of usage on a single charge. A battery protector module TP4056 is also provided to make the system rechargeable. The Inertial Measurement Unit sensor, ESP8266, battery, and the multiplexer will be placed within a 3D printed enclosure [29]. This enclosure containing all the circuity and components will be placed over the metatarsal region of the foot as it is the ideal position for gait measurement as well as to provide support to the components

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without interfering with any natural movement of the individual [30].

V. Result

The plantar pressure has been measured using force sensitive resistors which were placed in the insole of the foot. Using node MCU as the control unit, the data was computed and wirelessly transmitted to a LabVIEW dashboard. The different outputs that are displayed will be analyzed to study or predict different clinical conditions. The gait measurements were successfully identified using inertial measurement units. A detailed insole foot pressure and gait analysis were obtained from the Virtual Instrument.

This system was found to be working ideally with individuals in the height range of 145-160 cm and a weight range of 50-60 Kg. The graphical analysis from the individuals obtained is as listed below. The presented method and prototype are portable, energy efficient and economically feasible.

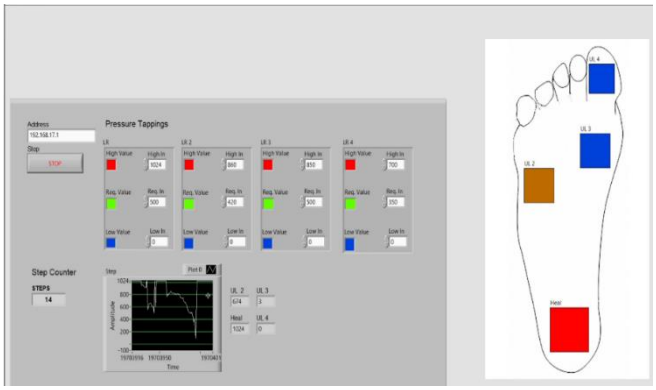

Fig 6: LabVIEW Output of the Foot Pressure Analysis

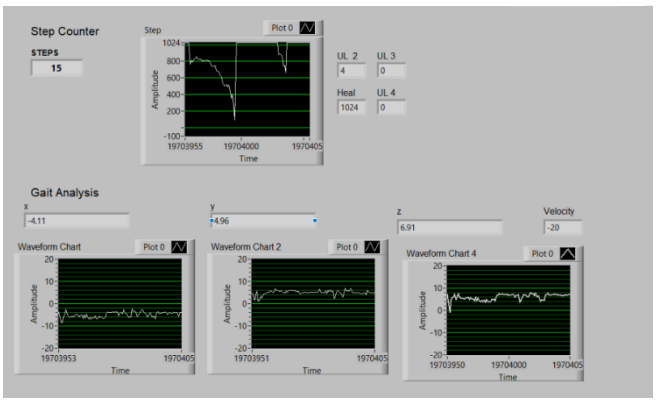

Fig 7: LabVIEW Output of the Gait Analysis

VI. Conclusion

This project deals with the measurement and analysis of foot pressure and gait characteristics of a human being. The proposed model helps in identifying diabetic foot ulcerations which might result in foot amputation under severe conditions. It also contributes to biomedical assessment after surgical procedures, intra-operative assessment, customized design of orthotics.

The proposed system works ideally for people with 50-70KG under all age categories. Development of diabetic ulcers can be noted in comparison with periodic measurement of plantar pressure and gait from the device. Apart from clinical applications it is widely used to study the mechanism of human posture and gait, on a long-term basis.

VII. Future Works

The future works for this project can be summarized into three phases. The first or initial phase includes building of a sufficiently large database in order to provide more credibility for the data gathered and the work completed.

The second phase includes the building of a completely finished product developed from the current prototype. The third and final phase would include the development of an Android application that receives signals from the wireless communication module. providing option to switch on alert notifications to nearby healthcare professionals. The device will also be tested with more patients having chronic diseases such as Parkinson's disease and comparisons with normal pressure distribution and gait will be monitored.

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